

Root–soil contact of field-grown winter wheat^α

M. van Noordwijk^a, D. Schoonderbeek^b and M.J. Kooistra^b

^aDLO Institute for Soil Fertility Research, P.O. Box 30003, NL 9750 RA Haren (Gr.),
The Netherlands

^bDLO Winand Staring Centre for Integrated Land, Soil and Water Research, P.O. Box 125, NL 6700
AC Wageningen, The Netherlands

(Received January 10, 1991; accepted after revision May 11, 1992)

ABSTRACT

Van Noordwijk, M., Schoonderbeek, D. and Kooistra, M.J., 1993. Root–soil contact of field-grown winter wheat. In: L. Brussaard and M.J. Kooistra (Editors), *Int. Workshop on Methods of Research on Soil Structure/Soil Biota Interrelationships*. Geoderma, 56: 277–286.

Roots following existing macropores and voids normally have only partial root–soil contact. Roots penetrating the soil matrix (creating new macropores) initially have complete root–soil contact. Partial root–soil contact may decrease the roots' effectiveness in taking up water and nutrients. If all roots have complete root–soil contact, however, aeration may be problematic unless roots have a high air-filled porosity. For field-grown winter wheat, root–soil contact was quantified from horizontally oriented thin sections at three depths, 15, 25 and 45 cm, respectively. One day prior to sampling, surface-connected pores were stained by infiltrating a methylene blue solution. Roots were observed microscopically using polarized light, and their diameter, roundness (indicating orientation) and degree of soil contact were measured with a Quantimet 970 image analyzer.

No relation was found between root–soil contact and root diameter or roundness. At 45 cm depth root–soil contact was less. For two fields, differing in soil organic matter content and current crop management, a different frequency distribution of root–soil contact was found in the plough layer. The percentage of roots with 100% root–soil contact was 65 and 37, that with 0% root–soil contact 5 and 14, respectively. For roots with partial root–soil contact the average degree of contact was about 60% in both cases. Average root–soil contact for the plough layer of the two fields was 84 and 66%, respectively. Roots without direct contact with the soil were growing mostly in surface-connected (blue stained) macropores. There was no difference in blue staining of the macropores with roots with 1–49% root–soil contact and those of the whole sample. Roots with 50–99% root–soil contact occurred mostly in relatively small, non-stained pores.

INTRODUCTION

Roots are interacting with soil structure in two ways: they may depend on pre-existing macropores and cracks (voids) to grow, and they may create new

Correspondence to: M. van Noordwijk, DLO Institute for Soil Fertility Research, P.O. Box 30003, NL 9750 RA Haren (Gr.), The Netherlands.

^αCommunication No. 45 of the Dutch Programme on Soil Ecology of Arable Farming Systems.

macropores by penetrating the soil matrix. After decay of the penetrating roots macroporosity can thus be increased. If roots penetrate the soil matrix they will initially have complete (100%) root-soil contact. If they follow pre-existing macropores or voids they will have no or partial contact with the soil, except when they completely fill a pre-existing macropore. Root-soil contact may change during the life of a root. Temporary root shrinkage during plant water stress or decay of epidermis and cortex leads to a decrease of root-soil contact. Secondary thickening of roots may increase root-soil contact.

The degree of root-soil contact is important for both aeration and uptake activities of the root. The chance that roots will obtain sufficient oxygen to survive periods of reduced soil aeration after heavy rainfall decreases with a higher degree of root-soil contact (De Willigen and Van Noordwijk, 1984, 1989). The ability to obtain water and nutrients from the soil matrix, however, increases with increasing root-soil contact (De Willigen and Van Noordwijk, 1987; Veen et al., 1992).

Root-soil contact may change with changing soil water status. Huck et al. (1970) described diurnal shrinkage-swelling cycles of roots under drought stress. Root shrinkage may lead to air gaps around the root and to increased resistance to water uptake (Faiz and Weatherley, 1982; Herkelrath et al., 1977). In these descriptions it is normally assumed that root-soil contact was complete in a wet soil and is reduced by root shrinkage. Actual measurements of the root-soil geometry are required to test these ideas. Thin-section preparation methods as described by Altemüller and Haag (1983) and Altemüller and Vorbach (1987) allow observation of root-soil geometry, including the role of root hairs. Samples are dehydrated by gradual acetone replacement, which is a time-consuming process.

Root-soil contact can also be quantified from thin-sections of the soil made by freeze-drying samples before impregnation of the soil with resin. Previous tests of this method on a pot experiment with maize at three soil porosities showed that roots do not shrink (at least not beyond the normal diurnal range) and that all root cross-sections are recognized (Van Noordwijk et al., 1992). In this pot experiment the average degree of root-soil contact was found to increase from 60 to 87% when soil porosity decreased from 60 to 44% (Kooistra et al., 1992). A study of root-soil contact for field-grown winter wheat was made as part of the Dutch Programme on Soil Ecology of Arable Farming Systems (Kooistra et al., 1989).

METHODS

Observations were made on June 7 1990 on two winter wheat fields at the Lovinkhoeve Experimental Farm in the Noordoostpolder (Marknesse, The Netherlands), with potato as the preceding crop. The soil is a calcareous silt loam with pH-KCL 7.5 and was reclaimed in 1942. At about 45 cm depth

there is a sandy layer of several cm thickness, with many shells, which restricts root development. Below that zone root length density normally increases again. A series of samples of 80 mm $\times$ 150 mm $\times$ 20 mm was taken from a vertical profile wall and horizontally oriented samples were taken at 15, 25 and 45 cm depth (the latter in the middle of the sandy layer), from a single soil pit in two fields: (1) field 16A with a relatively high soil organic matter content (2.7% in the top 20 cm) due to previous management, and currently under "integrated" management, i.e. reduced N-fertilization, reduced pesticide use and shallower soil tillage (12–15 cm cultivator in stead of 20–25 cm ploughing), (2) field 12B with a relatively low soil organic matter content (2.1% in the top 20 cm) due to previous management, and currently under "conventional" management.

Further data on the fields and the experiment are given by Kooistra et al. (1989). One day before the samples were taken, a methylene blue dye solution was infiltrated in the soil to stain surface-connected macropores and voids.

Thin sections, of 25 μ m thickness, were prepared after freeze-drying the samples as described by Van Noordwijk et al. (1992). Vertically oriented samples were used to measure macroporosity ($> 30 \mu$ m diameter). The horizontally oriented samples were scanned under a dissecting microscope, using crossed polarization filters below and above the sample to increase contrast between cell wall material and macropores. All roots were mapped on photographic enlargements (using the thin section as negative), and the root intensity (number of root intersections per cm^2) was measured. Quantitative analysis of roots and pore geometry was done with an image analyzer (Quantimet 970, Leica) at a magnification of $\times 213$ (length of one pixel 1.36 μ m), on 640 fields of view (0.94 mm^2 each) in the thin section. The primary root parameters measured were: number (N), area of cross-section (A), perimeter (P), largest (L) and smallest (S) Feret (caliper) diameter and percentage of root perimeter having direct root-soil contact. From the primary parameters a shape factor [$P^2/(4\pi A)$] was derived indicating the "roundness" of the roots. For a transverse section of a cylindrical root roundness is close to 1.0; for an elliptical shape of a root intersecting the plane of observation at a smaller angle, higher values (at least up to 5.0) are found. In a population of circles and ellipses the roundness factor thus indicates the orientation of roots. Estimates of the angle of interception from $\arcsin(S/L)$ may, however, contain a serious bias (Van Noordwijk et al., 1992). Within the population of roots on a single thin-section, a regression analysis was performed to test whether or not root-soil contact is related to root diameter or roundness. Comparisons of proportions of roots in four root-soil contact classes (0, 1–49, 50–99, 100%) between thin sections were based on a chi-square test.

A series of auger samples (7 cm diameter; 12 per field) was taken on the same day as the soil samples for the thin sections. The auger samples were

washed over a fine sieve (0.3 mm mesh), root length was determined with the line intersect method (Tennant, 1975) and diameter was measured on 20 randomly chosen roots per sample under a dissecting microscope.

RESULTS

Root length density and root diameter in the auger samples and root intensity on the thin sections are given in Table 1. The average root diameter on the thin sections was 0.178 mm; the average for the equivalent auger samples, weighted for root length per sample, was 0.203 mm. As the frequency distribution of root diameters on the thin section was skewed, a logarithmic transformation was used and the few roots of more than 2 mm diameter were discarded. Expressed this way, the average root diameter on the thin sections increased significantly with depth, from 0.161 to 0.208 and 0.242 mm for 15, 25 and 45 cm depth, respectively. No difference in root diameter between the two fields was found.

The average roundness of roots was 1.75. No effects of depth or field on roundness, a rough indication of root orientation, were found. Roundness and root diameter were not correlated either.

Figure 1 presents two of the thin sections showing the two main positions of roots in soil: growing in macropores along aggregate surfaces (A) and penetrating the soil matrix (B). The first group has a partial, the second a complete root-soil contact. Figure 2 shows details of root-soil contact, varying from 0 to 100%.

Figure 3 shows the frequency distribution of root-soil contact observed in

TABLE 1

Root diameter (Diam) and root length density (L_{rv}) observed in auger samples and root intensity (N) observed on horizontally oriented thin section samples for two fields, 12B and 16A

Depth, cm	Diam (mm)		L_{rv} (cm/cm ³)		N (cm ⁻²)	
	12B	16A	12B	16A	12B	16A
0- 5	0.203	0.223	13.2	18.4		
5- 10	0.199	0.197	7.87	10.1		
10- 20	0.193	0.188	3.85	4.98	4.90	4.25
20- 30	0.206	0.227	2.94	3.22	2.85	1.86
30- 40	0.233	0.241	1.49	1.80		
40- 50	0.220	0.239	1.27	1.06	0.81	0.28
50- 60	0.213	0.221	1.44	1.37		
60- 70	0.191	0.217	1.20	1.74		
70- 80	0.187	0.219	0.84	1.47		
80- 90	0.196	0.227	0.28	0.95		
90-100	0.199	0.220	0.12	0.29		

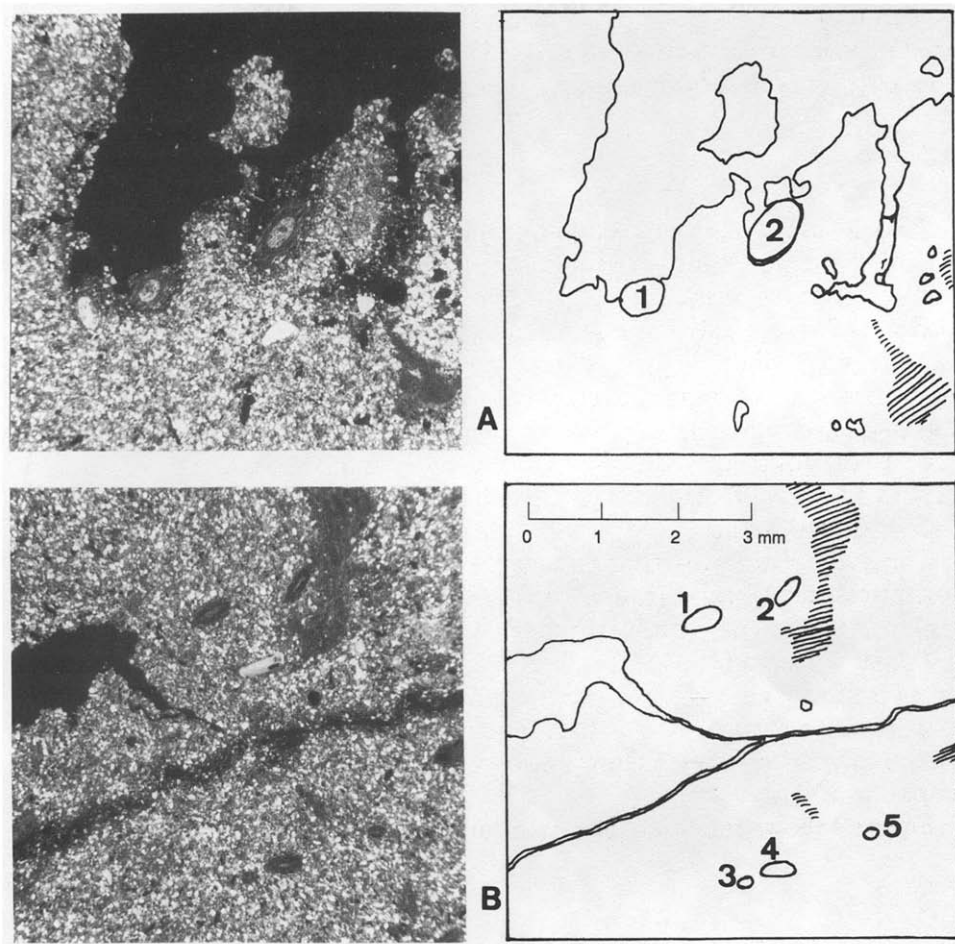


Fig. 1. Overview of two main positions of roots in soil. (A) Two roots growing in a void, along an aggregate wall, with partial root-soil contact. (B) Five roots penetrating the soil matrix, with 100% root-soil contact. Hatched areas on the sketches indicate organic matter accumulations; black areas on the photographs are voids (macropores and cracks).

six thin-section samples. Linear regression analysis showed that no relation existed between root-soil contact and root diameter (only 7% of variation accounted for) or roundness (no variation accounted for). A chi-square test of homogeneity of the frequency distributions indicated ($0.05 < p < 0.10$) a difference between the three depth layers, when results for the two fields were pooled. For a combination of the two samples in the plough layer, a chi-square test indicated a significant ($p < 0.05$) difference between the two fields in frequency distribution of root-soil contact. In field 16A fewer roots with complete root-soil contact (forming their own channel) were found, and for the

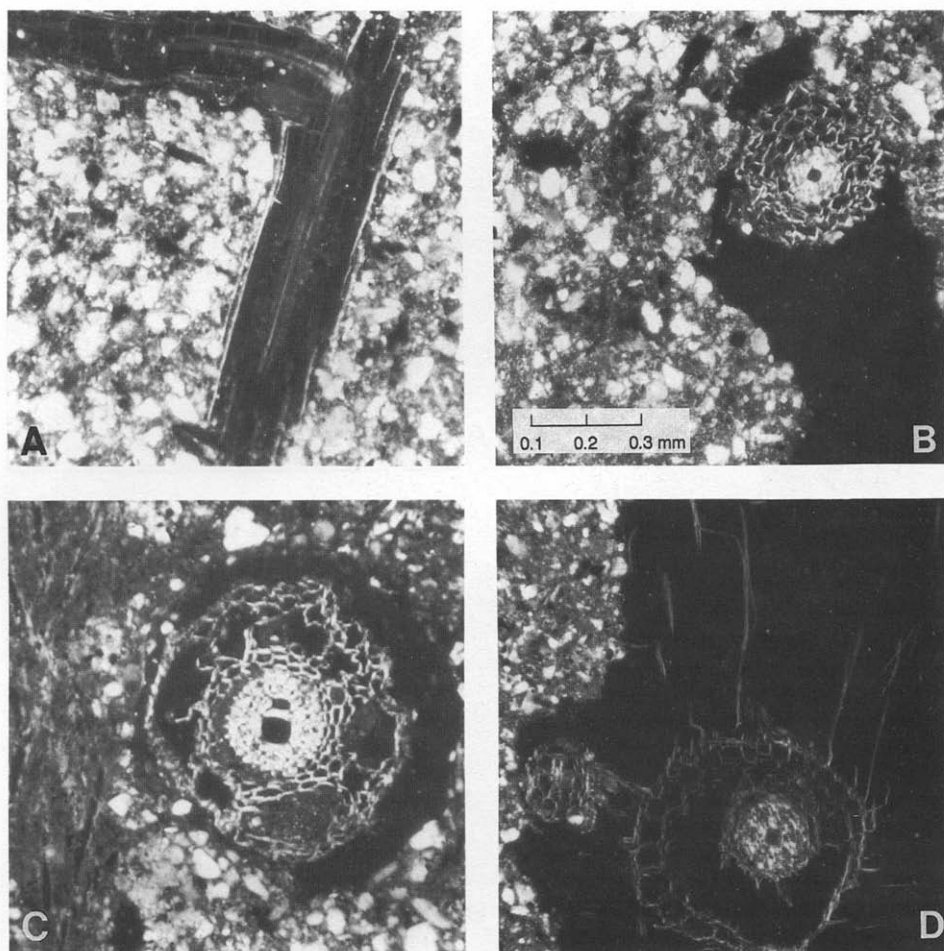


Fig. 2. Details of root-soil contact of wheat as observed on thin sections. (A) Longitudinal section of branching root with 100% root-soil contact. (B) Transverse section of root growing in a void with direct soil contact on two sides, root penetrating the soil matrix. (C) Transverse section of root with partial contact; note cortical aerenchyma. (D) Transverse section of root without direct soil contact, but with root hairs extending towards the soil matrix and a branch root with partial contact.

roots growing in pre-existing macropores a larger part had no direct contact with the soil. In the latter group root hairs may still establish partial contact with the soil, but only on a fraction of the roots without soil contact root hairs were seen. Table 2 summarizes root-soil contact data for the plough layer at the two fields. Average root-soil contact differed significantly between the two fields.

Part of the macropores in which roots were found were stained with meth-

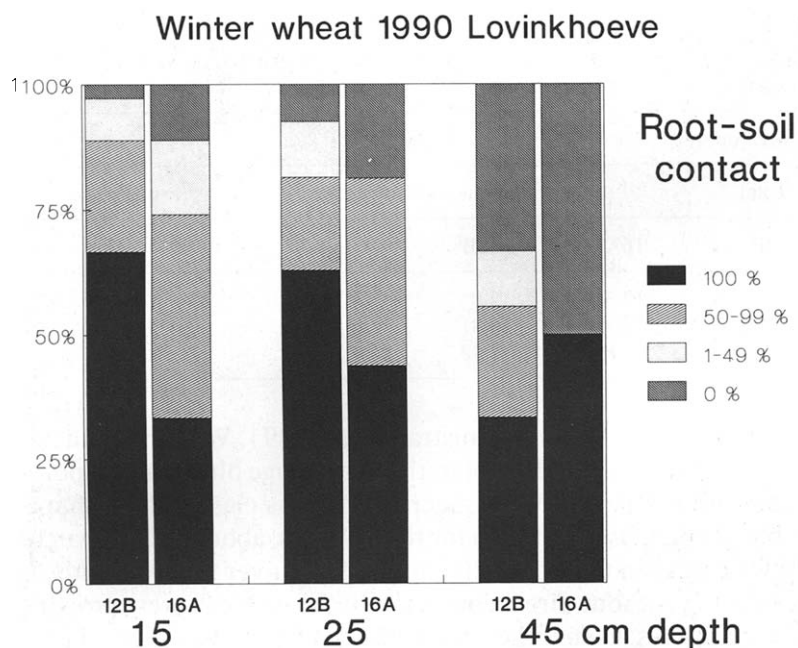


Fig. 3. Frequency distribution of root-soil contact of wheat in horizontal thin-section samples taken at three depths in two fields of the Lovinkhoeve experiment. Number of roots observed in the 640 fields of view (0.94 mm^2 each) on six thin sections: 36, 27, 27, 16, 9 and 4, respectively.

TABLE 2

Root-soil contact of wheat for horizontal thin sections at 15 and 25 cm depth for two fields, 12B and 16A

Root-soil contact class (%)	Percentage of roots		Average contact	
	12B <i>N</i> = 63	16A <i>N</i> = 43	12B	16A
0	4.8	14.0	0	0
1-99	30.2	48.8	61.7	58.6
100	65.1	37.2	100	100
Total	100	100	83.7	65.8

ylene blue and are thus considered to be surface-connective. Table 3 shows the macroporosity (diameter $> 30 \mu\text{m}$) for vertically oriented thin-section samples, taken close to the horizontally oriented ones used for evaluating root-soil contact. The macroporosity on the thin sections is relatively high, especially on field 16A, compared to previous measurements (Kooistra et al., 1989). About 30% of the macropores in the upper 20 cm of the soil is blue

TABLE 3

Macroporosity ($> 30 \mu\text{m}$ diameter) in thin-section samples for three depth zones of vertically oriented thin-section samples of field 12B and 16A

Depth (cm)	Macroporosity (% v/v)					
	Total		Blue stained		Percentage blue	
	12B	16A	12B	16A	12B	16A
01-09	6.26	19.35	2.04	6.61	33	34
11-19	14.63	23.15	6.81	5.06	46	22
21-29	4.65	6.80	0.79	0.63	17	9

stained, similar to results for 1985 (Kooistra et al., 1989). When the macropores are classified by equivalent diameter, the percentage blue-stained pores tends to increase with diameter. When macroporosity is classified by shape, the percentage blue-stained is about 20% for round pores, about 25% for vughs and about 40% for cracks in the upper 20 cm, averaged over the two fields.

Of the roots growing without direct contact with the soil 56% were growing in blue-stained macropores; even if they occur mainly in relatively large cracks, they appear to prefer blue-stained and thus surface-connective cracks. Of the roots with 1-49% and 50-99% contact 20 and 7%, respectively, were growing in blue-stained macropores. Roots with 100% soil contact were not stained.

DISCUSSION

The average root diameter on the thin sections was 10% less than the diameter of roots washed from auger samples. This might be caused by root shrinkage during sample preparation, or by losses of fine rootlets when washing or by a different criterion for separating live and dead roots. Root shrinkage is probably not a major cause as we observed many roots with 100% root-soil contact. When the thin-section method was tested on a maize pot experiment, with soil without root remains, the root intensity observed (N, cm^{-2}) agreed with the root length density ($L_{rv}, \text{cm}/\text{cm}^3$) from washed replicate samples, following the theoretical relation $L_{rv} = 2N$ (Van Noordwijk et al., 1992). For the current samples we found $L_{rv} = 0.97N$ on field 12B and $L_{rv} = 1.3N$ on field 16A, averaged over 15 and 25 cm depth. This could be due to the fact that the thin section samples had a higher than average root density (the data fall within the range of root densities on individual auger samples) or to the fact that part of the roots observed on the thin section would be discarded as dead roots in the washed samples. In discussing the data the possibility of the latter explanation should be kept in mind.

Roughly two thirds of the roots formed their own root channels on the thin

sections of field 12B, as derived from the frequency of 100% root-soil contact, and only one third on 16A. The considerably higher macroporosity of the plough-layer samples of field 16A, with its higher soil organic matter content, suggest that more roots could follow pre-existing pores on that field, while roots on field 12B had to penetrate the soil matrix. Roots occupied only a small fraction (1–5%) of the available, blue-stained macropore space, however, even on field 12B. The small size of the thin section samples, taken from one soil pit in each field, makes it impossible to extrapolate from the samples to the field scale. Because cropping history and current management are confounded in the comparison of the two fields (Kooistra et al., 1989), no conclusions on the influence of either of them can be made. The differences found in root-soil contact are primarily related to a difference in macroporosity, not to specific causes of this difference in macroporosity.

The average values for root-soil contact in this study are within the range found earlier for maize (Kooistra et al., 1992). Root hairs which may make up for incomplete root-soil contact were seen only rarely, although some examples show that they survive sample preparation without apparent damage. As noted in a pot experiment (Kooistra et al., 1992), roots growing in existing macropores and cracks have more contact with the soil than would be expected for a random positioning in the macropore space. Roots can apparently find and follow the walls of the macropores and cracks.

Roots without direct contact with the soil were growing mostly in surface-connected (blue-stained) macropores. Roots with 1–49% root-soil contact occurred in macropores with an average fraction of blue staining, while roots with 50–99% root-soil contact occurred mostly in relatively small, non-stained pores. Roots penetrating the soil matrix (100% root-soil contact) probably depend on internal aeration through cortical air spaces. A certain part of each root axis should have air contact to serve as “breathing sections”. The observation that roots with complete air contact grow mostly in surface-connected pores agrees remarkably well with this interpretation.

REFERENCES

- Altemüller, H.J. and Haag, Th., 1983. Mikroskopische Untersuchungen an Maiswurzeln im ungestörten Bodenverband. *Kali-Briefe* (Büntehof), 16: 349–363.
- Altemüller, H.J. and Vorbach, A., 1987. Veränderungen des Bodengefüges durch Wurzelwachstum von Maispflanzen. *Mitt. Dtsch. Bodenkundl. Gesellsch.*, 55: 93–98.
- De Willigen, P. and Van Noordwijk, M., 1984. Mathematical models on diffusion of oxygen to and within plant roots, with special emphasis on effects of soil root contact. I. Derivation of models, II. Applications. *Plant Soil*, 77: 215–241.
- De Willigen, P. and Van Noordwijk, M., 1987. Roots, plant production and nutrient use efficiency. Doctoral Thesis, Wageningen Agricultural University, 282 pp.
- De Willigen, P. and Van Noordwijk, M., 1989. Model calculations on the relative importance

- of internal longitudinal diffusion for aeration of roots of non-wetland plants. *Plant Soil*, 113: 111–119.
- Faiz, S.M.A. and Weatherley, P.E., 1982. Root contraction in transpiring plants. *New Phytol.* 81: 19–28.
- Herkelrath, W.N., Miller, E.E. and Gardner, W.R., 1977. Water uptake by plants. II. The root contact model. *Soil Sci. Soc. Am. Proc.*, 41: 1039–1043.
- Huck, M.G., Klepper, B. and Taylor, H.M., 1970. Diurnal variations in root diameter. *Plant Physiol.*, 45: 529–30.
- Kooistra, M.J., Lebbink, G. and Brussaard, L., 1989. The Dutch programme on soil ecology of arable systems. 2. Geogenesis, agricultural history, field site characteristics and present farming systems at the Lovinkhoeve experimental farm. *Agric. Ecosyst. Environ.*, 27: 361–387.
- Kooistra, M.J., Schoonderbeek, D., Boone, F.R., Veen, B.W. and Van Noordwijk, M., 1992. Root–soil contact of maize, as measured by a thin-section technique. II. Effects of soil compaction. *Plant Soil*, 139: 119–129.
- Tennant, D., 1975. A test of a modified line intersect method of estimating root length. *J. Ecol.*, 63: 995–1001.
- Van Noordwijk, M., Kooistra, M.J., Boone, F.R., Veen, B.W. and Schoonderbeek, D., 1992. Root–soil contact of maize, as measured by a thin-section technique. I. Validity of the method. *Plant Soil*, 139: 109–118.
- Veen, B.W., Van Noordwijk, M., De Willigen, P., Boone, F.R., Kooistra, M.J., 1992. Root–soil contact of maize, as measured by a thin-section technique. III. Effects on shoot growth, nitrate and water uptake efficiency. *Plant Soil*, 139: 131–138.